Assignment-3.4

TASK-1

Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.

Step-by-Step RAM Reseating Procedure

- **Power Down and Disconnect:** Shut down the computer completely. Unplug the power cable from the wall outlet and disconnect all external peripherals.
- **Discharge Residual Power:** Press and hold the computer's power button for 10 to 15 seconds to drain any residual electrical charge stored in the capacitors.
- **Open the Case:** Remove the side panel (for desktops) or the bottom access panel (for laptops) using the appropriate screwdriver.
- **Ground Yourself:** Touch a bare metal part of the computer chassis or wear an anti-static wrist strap to prevent Electrostatic Discharge (ESD) damage to the components.
- **Locate the RAM Slots:** Identify the memory modules on the motherboard. They are typically long, slender modules held in place by clips on either side.
- **Remove the RAM Module:** Press down firmly on the plastic retaining clips at both ends of the RAM slot. The module will pop up slightly. Grip the module by its edges and pull it straight out. Do not touch the gold metal contacts.
- **Inspect and Clean (Optional):** Check the slot for dust. If necessary, blow it out with compressed air.

- **Re-insert the RAM Module:** Align the notch on the bottom edge of the RAM stick with the ridge in the memory slot to ensure correct orientation. Push the module straight down firmly into the slot until the retaining clips snap back into place automatically.
- **Close and Power On:** Replace the computer panel, reconnect the power cable, and boot up the system.

Observed Issues and Troubleshooting During the Process

- **Misalignment:** The RAM module is keyed with a physical notch. Trying to insert it backwards will prevent it from seating and can damage the slot if forced.
- **Incomplete Seating:** If the clips do not snap into place completely, the computer will likely fail to boot, resulting in a black screen and a series of audible beep codes (or motherboard diagnostic LED lights) indicating a memory error.
- **Stiff Retention Clips:** On older motherboards, the plastic clips can become brittle or resistant to opening and closing, requiring careful pressure to avoid breaking them.

TASK-2

Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix.

1. Identify Symptoms & Isolate the Display

Before opening the PC case, determine if the issue is actually the GPU or simply a bad peripheral connection.

- Check the cables**:** Ensure your HDMI or DisplayPort cable is firmly plugged directly into the GPU, not the motherboard.
- Test the monitor**:** Connect the PC to a different monitor or a TV, or swap the display cable.
- Note the symptoms**:** Document exactly what is happening to help narrow down the cause:
    - No signal / Black screen
    - Screen artifacts (lines, dots, or weird colors on screen)
    - Frequent game crashes or "Driver timed out" errors
    - BSOD (Blue Screen of Death) related to graphics drivers (nvlddmkm.sys, amdkmdag.sys)

2. Perform a Clean Driver Installation

Software conflicts and corrupted graphics drivers frequently mimic hardware failures.

- Boot into Safe Mode**:** Boot Windows into Safe Mode to prevent corrupted drivers from loading during the test.
- Run DDU (Display Driver Uninstaller): Use DDU to completely wipe all traces of past and present graphics drivers from your system.

- Install fresh drivers**:** Restart the PC normally, download the latest stable driver directly from the official manufacturer website (NVIDIA, AMD, or Intel), and perform a clean installation.

3. Inspect Physical Connections & Power

Loose hardware or inadequate power delivery can prevent a GPU from functioning correctly under load.

- Reseat the GPU**:** Turn off the PC, unplug the power cord, open the case, and remove the GPU. Inspect the PCIe slot for dust, then firmly click the GPU back into the primary PCIe x16 slot.
- Check PCIe power cables**:** Ensure the 6-pin, 8-pin, or 12VHPWR power connectors are completely pushed in and clicked into place on both the GPU and the power supply (PSU) side.
- Verify PSU capacity**:** Ensure your power supply meets the minimum wattage requirements for your specific GPU model.

4. Test with Integrated Graphics or a Backup GPU

Hardware cross-testing is the definitive way to prove whether the GPU itself is broken.

- Switch to Integrated Graphics (iGPU): If your CPU has built-in graphics, remove the dedicated GPU entirely and plug your monitor into the motherboard. If the PC boots and displays perfectly, the dedicated GPU is likely the issue.
- Test in a different PC**:** If possible, install the suspect GPU into a friend's known-working computer, or try a known-working backup GPU in your own PC.

5. Stress Test & Confirm the Fix

Once you have taken corrective action, you must stress test the system to confirm that the stability issue is permanently resolved.

- Run a synthetic benchmark**:** Run demanding loops in software like 3DMark*, FurMark*, or Superposition for at least 30 to 60 minutes.
- Monitor vitals**:** Use hardware monitoring software (like HWMonitor or MSI Afterburner) to verify that GPU core and hotspot temperatures stay within safe operational limits (typically under 85°C).
- Real-world gaming**:** Launch a graphically intensive game that previously caused crashes and play for an hour to confirm the system remains stable.

TASK-3

Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.

Phase 1: Unplugging the Drive (System Running)

SATA Drive (HDD/SSD)

- **The Action:** Disconnecting the SATA data/power cable while the OS is actively running.
- **Error Messages (OS Level):**
    - *Windows:* A Blue Screen of Death (BSOD) appears with the stop code **CRITICAL_PROCESS_DIED** or **KERNEL_DATA_INPAGE_ERROR**.
    - *Linux:* The terminal or system logs (dmesg) flood with errors: **blk_update_request: I/O error, dev sda** followed by **SATA link down**.

NVMe Drive (M.2 PCIe)

- **The Action:** Removing an NVMe drive from its M.2 slot while running (Note: NVMe slots are generally *not* designed for hot-plugging unless using specialized U.2/U.3 enterprise bays).
- **Error Messages (OS Level):** The system typically freezes completely before a BSOD can even write to a log. If it manages to trigger a BSOD, it displays **WHEA_UNCORRECTABLE_ERROR** (indicating a critical hardware failure).

Phase 2: Rebooting Before Reconnecting (The "Drive Missing" State)

When the system restarts *without* the drive attached, it fails to clear the Power-On Self-Test (POST) sequence normally.

BIOS/UEFI Indicators:

- **Main Dashboard:** Under "Storage Information," the affected port (e.g., *SATA Port 1*) or slot (e.g., *M.2_1 Slot*) reads Empty or Not Detected.
- **Boot Priority List:** The drive's name (e.g., *Samsung SSD 990 PRO* or *WDC WD10EZEX*) completely disappears from the bootable devices list.

Black Screen Error Messages:

Because no bootloader can be found, the motherboard displays one of the following classic firmware errors:

- Reboot and Select proper Boot device or Insert Boot Media in selected Boot device and press a key
- No Bootable Device Found.
- *On older systems/Intel boards:* **PXE-E61: Media test failure, check cable** (The system is trying to boot from the network because local storage failed).

Phase 3: Reconnecting the Drive & Rebooting

- **The Action:** Turning off the system, firmly reseating the SATA or NVMe drive, and powering it back on.

BIOS/UEFI Indicators (After Reconnection):

- **Successful POST:** The system successfully passes the initial hardware check without pausing on a black screen.
- **Storage Map:** In the BIOS menu, the drive reappears under its respective port/slot with its full factory model name and capacity intact.
- **Boot Order:** The **Windows Boot Manager** or **GRUB** entry automatically moves back to the top of the boot priority list.

TASK-4

Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.

When a PC randomly restarts without a Blue Screen of Death (BSOD), RAM is the first component you should check.

1. Power off the PC and unplug it from the power supply.

2. Open the PC case and locate the RAM modules on the motherboard.

3. Check the RAM physically for dust, damage, or improper seating.

4. Reseat the RAM by removing it and firmly reinstalling it until the retaining clips lock.

5. Test one RAM module at a time if multiple modules are installed. This can help identify a faulty stick.

6. Try different RAM slots if the motherboard supports them.

7. Run a memory diagnostic test such as Windows Memory Diagnostic to check for memory errors.

8. Confirm the fix Use the PC normally or run a memory-intensive task and check whether the random restarts have stopped.